

Ideation Phase Brainstorm & Idea Prioritization

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Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization:

Brainstorming provides an open environment in which every team member can contribute ideas towards solving a problem. Volume of ideas is encouraged over early judgment, unusual ideas are welcomed and built upon, and all members collaborate to develop a wide range of possible solutions.

This document records the brainstorming session carried out by the five-member team for EduGenie: Google Gemini Powered Learning Assistant. It follows three steps: selecting the problem statements, listing and grouping ideas, and prioritizing the idea groups.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

A. Team Gathering

The team has five members. Each member took a focus area so that all identified student problems were discussed from the student's point of view.

Team Member	Name	Brainstorming Focus Area
Member 1	Mohammed Yasin	Concept Understanding
Member 2	Gokulnath	Personalized Explanation and Clarification
Member 3	Mathavan	Structured Learning
Member 4	Jayapriyan	Revision and Summarization
Member 5	Dinesh Kumar	Practice and Assessment

B. Set the Goal

The objective of the brainstorming session was to generate possible solution ideas that can help students:

- understand difficult concepts more easily
- clarify their doubts
- follow a structured learning progression
- review learning content efficiently
- practice and assess their understanding

C. Define Your Problem Statement

The team discussed the following three core problem statements, framed as “How might we...?” questions, which became the focus of the brainstorm.

ID	Core Problem Statement	How Might We...?
PS-1	Difficulty Understanding Complex Academic Concepts	How might we help students understand difficult academic concepts more easily and clear their doubts?
PS-2	Difficulty Following a Structured Learning Progression	How might we help students know what to learn first and what to learn next in a subject?
PS-3	Difficulty Efficiently Reviewing and Practicing Learning Content	How might we help students revise lengthy material efficiently and check their understanding?

Key Rules of Brainstorming	
Stay on topic.	Encourage wild ideas.
Defer judgment.	Listen to others.
Go for volume.	If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

A. Brainstorm

Each member wrote down ideas that address the three problem statements. Each note below represents one idea contributed by that member.

Member 1	Member 2	Member 3	Member 4	Member 5
Concept Understanding	Personalized Explanation and Clarification	Structured Learning	Revision and Summarization	Practice and Assessment
Simplify complex academic concepts using easy language.	Provide explanations according to the student's learning level.	Recommend the next topic to learn.	Summarize lengthy learning content.	Generate quizzes from learning content.
Provide step-by-step explanations for difficult topics.	Provide beginner-friendly explanations.	Organize topics from basic to advanced.	Extract important points.	Generate topic-based questions.
Explain difficult academic terminology in simpler words.	Provide more detailed explanations when required.	Create a structured learning path.	Generate concise revision notes.	Generate multiple-choice questions.
Provide examples along with concept explanations.	Explain the same concept in different ways.	Suggest prerequisite topics.	Convert lengthy explanations into easy-to-review content.	Identify areas that require further revision.
Allow students to ask questions when they do not understand a concept.	Provide follow-up clarification for misunderstood concepts.	Recommend topics based on the student's current learning stage.	Highlight key concepts for quick revision.	Provide feedback after answering questions.

B. Group Ideas

Similar and related ideas were shared and clustered into five idea groups. Each group is given a short label and a purpose. The member who contributed each idea is shown in brackets.

Group 1: Concept Simplification & Explanation

Purpose: Help students understand difficult concepts through clear, simple, step-by-step and level-appropriate explanations.

- Simplify complex academic concepts using easy language. (Member 1)
- Provide step-by-step explanations for difficult topics. (Member 1)
- Explain difficult academic terminology in simpler words. (Member 1)
- Provide examples along with concept explanations. (Member 1)
- Provide explanations according to the student's learning level. (Member 2)
- Provide beginner-friendly explanations. (Member 2)
- Provide more detailed explanations when required. (Member 2)
- Explain the same concept in different ways. (Member 2)

Group 2: Interactive Doubt Clarification

Purpose: Allow students to ask questions and receive additional explanations when they do not understand a concept. This is a supporting capability for Problem 1.

- Allow students to ask questions when they do not understand a concept. *(Member 1)*
- Provide follow-up clarification for misunderstood concepts. *(Member 2)*

Group 3: Personalized Learning Progression

Purpose: Guide students from foundational concepts toward advanced topics in a structured sequence.

- Recommend the next topic to learn. *(Member 3)*
- Organize topics from basic to advanced. *(Member 3)*
- Create a structured learning path. *(Member 3)*
- Suggest prerequisite topics. *(Member 3)*
- Recommend topics based on the student's current learning stage. *(Member 3)*

Group 4: Efficient Revision & Summarization

Purpose: Help students efficiently review lengthy learning material by focusing on important information and concise summaries.

- Summarize lengthy learning content. *(Member 4)*
- Extract important points. *(Member 4)*
- Generate concise revision notes. *(Member 4)*
- Convert lengthy explanations into easy-to-review content. *(Member 4)*
- Highlight key concepts for quick revision. *(Member 4)*

Group 5: Practice & Knowledge Assessment

Purpose: Help students practice learned concepts and assess their understanding through generated questions and quizzes.

- Generate quizzes from learning content. *(Member 5)*
- Generate topic-based questions. *(Member 5)*
- Generate multiple-choice questions. *(Member 5)*
- Identify areas that require further revision. *(Member 5)*
- Provide feedback after answering questions. *(Member 5)*

C. Idea Group → Problem → Capability Mapping

Each idea group supports one of the three core problems. Interactive Doubt Clarification is a supporting capability for Problem 1 and is not treated as a separate problem.

Idea Group	Core Problem	EduGenie Capability
Concept Simplification & Explanation	Problem 1	Simplified Concept Explanation
Interactive Doubt Clarification	Problem 1	Question Answering / Doubt Clarification
Personalized Learning Progression	Problem 2	Learning Recommendations / Structured Learning Path
Efficient Revision & Summarization	Problem 3	Summarization
Practice & Knowledge Assessment	Problem 3	Quiz Generation / Knowledge Assessment

The overall relationship between the student problems, student needs and EduGenie capabilities is summarised below.

Student Problem	Student Need	Idea Groups	EduGenie Capability
PS-1: Difficulty Understanding Complex Academic Concepts	Understand concepts clearly and clear doubts	Concept Simplification & Explanation + Interactive Doubt Clarification	EduGenie Explanation + Q&A
PS-2: Difficulty Following a Structured Learning Progression	Know what to learn first and what to learn next	Personalized Learning Progression	EduGenie Learning Recommendations / Learning Path
PS-3: Difficulty Efficiently Reviewing and Practicing Learning Content	Revise efficiently and check understanding	Efficient Revision & Summarization + Practice & Knowledge Assessment	EduGenie Summarization + Quiz Generation

Step-3: Idea Prioritization

The team placed the five idea groups on an Importance vs Feasibility grid to decide which ideas to take forward. **Importance** asks which group would have the most positive impact for students. **Feasibility** asks which group is more practical to implement in terms of effort, time and complexity. Placement is qualitative; no numerical scores are used.

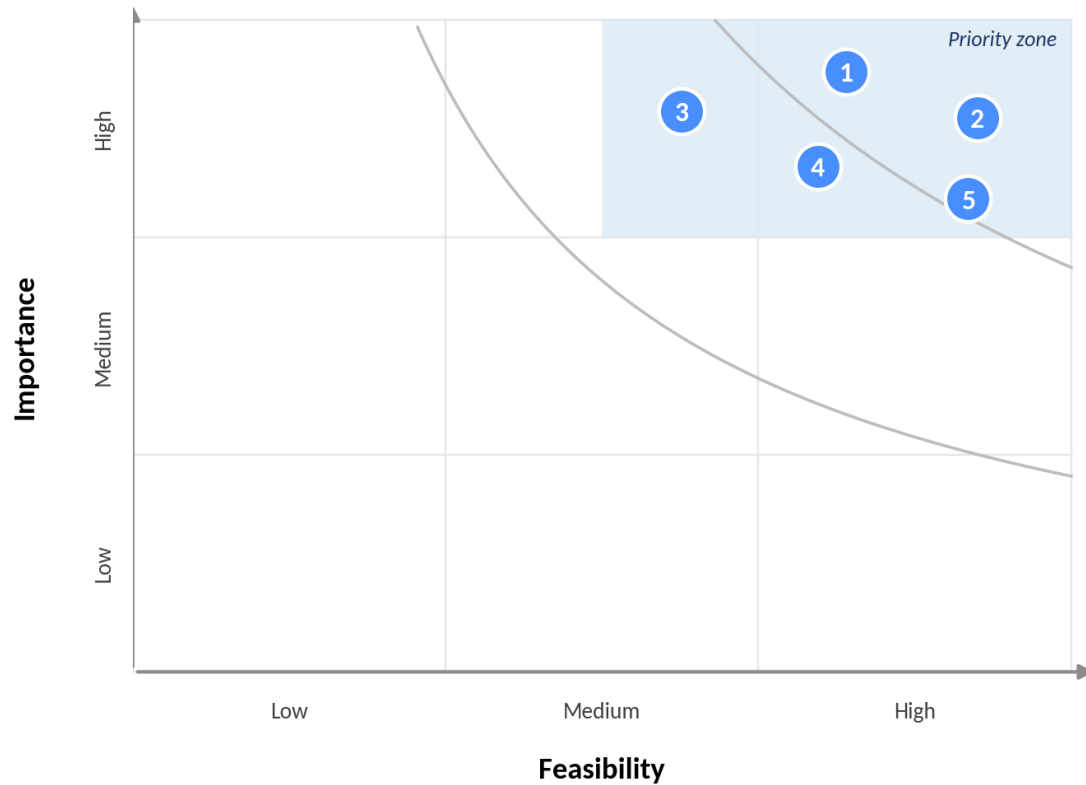


Figure 1: Importance vs Feasibility matrix for the EduGenie idea groups

No.	Idea Group
1	Concept Simplification & Explanation
2	Interactive Doubt Clarification
3	Personalized Learning Progression
4	Efficient Revision & Summarization
5	Practice & Knowledge Assessment

Prioritization Summary

Idea Group	Importance	Feasibility	Prioritization Rationale
Concept Simplification & Explanation	High	High	Directly addresses students' difficulty understanding complex concepts and can be implemented using the project's AI-based explanation capability.
Interactive Doubt Clarification	High	High	Supports students who need immediate clarification and complements the simplified explanation capability.
Personalized Learning Progression	High	Medium-High	Helps students follow a structured learning sequence, while requiring more contextual understanding of learning progression.
Efficient Revision & Summarization	High	High	Helps students reduce lengthy content into concise revision material.
Practice & Knowledge Assessment	High	High	Enables students to practice concepts and assess their understanding through generated questions and quizzes.

Final Brainstorming Outcome

The brainstorming activity resulted in five major solution directions. These are solution directions that support the three core student problems; they are not separate problem statements.

No.	Solution Direction	Core Problem	EduGenie Capability
1	Clear and Simplified Concept Understanding	Problem 1	Simplified Concept Explanation
2	Interactive Doubt Clarification	Problem 1	Question Answering / Doubt Clarification
3	Structured Learning Progression	Problem 2	Learning Recommendations / Structured Learning Path
4	Efficient Content Revision	Problem 3	Summarization
5	Learning Practice and Assessment	Problem 3	Quiz Generation / Knowledge Assessment